INCH-POUND

MIL-DTL-50714C (OS)
27 June 2004
SUPERSEDING
MIL-DTL-50714B (OS)
1 December 1995

DETAIL SPECIFICATION

INITIATOR, PROPELLANT ACTUATED, DELAY, M113 ASSEMBLY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M113 Propellant Actuated Delay Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550	Ammunition, General Specification For
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to pennam@ih.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

MIL-D-50727	Delay Element Assembly
MIL-H-25579	Hose Assembly Tetrafluoroethylene, High Temperature Power Plant, Aircraft
MIL-DTL-50374	Initiator, Propellant Actuated, M104 Assembly
MIL-DTL-50375	Initiator, Propellant Actuated, M104 Metal Parts Assembly
MIL-P-19834	Plates, Identification, Metal Foil, Adhesive-Backed

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

DL838AS163	Test Fixture, 6.5 Cubic Inch Closed Bomb
9280892	Box, Packing, Ammunition For Initiation, Delay (Gas - Subminiature)
9280893	Box, Fiberboard (Outer), Alternative, Packing Ammunition for Initiator, Delay (Gas - Subminiature)
DL10551446	Initiator, Propellant Actuated, M104 Assembly
11731474	Initiator, Propellant Actuated, Delay, M113 Assembly

U.S. Air Force

TPO F00-138-3770	Initiators

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following document form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Inspection, Radiographic

(Copies of this document are available online at http://www.astm.org/cgi-bin/SoftCart.exe/STORE/standardsearch.shtml?L+mystore+afpt6301+1062008944 or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample of bellows actuators shall be subjected to first article inspection (see 4.4 and 6.3)

3.2 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.3 Primary components. For the purpose of this specification, the delay element (Drawing 11738753) and the M104 Propellant Actuated Initiator (Drawing 10551446) are defined as primary components. Only primary components from a single lot shall be used in a production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

3.4 Ignition device. The ignition device shall conform to the applicable drawing and referenced specification. Prior to assembly into a retainer, the ignition device shall be visually examined and X-rayed.

3.4.1 Sensitivity test. The PVU-1/A Ignition Device, when assembled and crimped into an ignition device retainer, part no. 11731620, shall fire when drop tested at an energy level of 26.0 inch ounces and shall not fire when drop tested at an energy level of 4.0 inch ounces.

3.5 Breakaway torque. Each threaded joint of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation using the procedure of 4.5.3.

3.6 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction using the procedure of 4.5.4.

3.7 Functional.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

3.7.1 Ballistics. Initiators loaded with accepted pyrotechnic components (II-A pellets, PVU-1/A Ignition Device, and delay element) and temperature conditioned shall produce values specified in table I. Pressure values shall be recorded in a 6.5-in^3 closed bomb, Drawing 838AS163.

TABLE I. Ballistic values.

Temperature	Pressure, psi (min)	Ignition delay, sec (max) (see 6.3)
-65 to -70°F	300	0.300 ± 0.150
65 to 75°°F	300	0.300 ± 0.050
200 to 205? F	300	0.300 ± 0.150

3.7.1.1 Initiation. The initiator shall be fired by a source delivering an output energy equal to that of an M3A2 Initiator at the end of 15 feet of aircraft hose, size no. 4 in accordance with MS28741 at 70 ± 5°F.

3.7.2 No-fire. The initiator shall not fail to fire.

3.7.3 Mechanical failure. The initiator shall function without mechanical failure (see 6.2).

3.8 Leak rate. The assembled initiator, less the firing mechanism assembly (part no. 11731492) and the mounting bracket (part no. 11731624), shall withstand a 14 psi internal-external pressure differential with a maximum permissible leak rate of 1 x 10^5 cubic centimeters per second.

3.9 Workmanship. Workmanship shall be as specified on the applicable drawings, referenced specifications, and the following:

3.9.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any other metal defect.

3.9.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion, or other foreign matter.

3.10 Mechanical properties. Mechanical properties required by the drawing shall be uniform throughout the part within specified tolerances.

3.11 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

3.12 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour.

3.13 Radiographic examination. Each ignition device assembly, retainer, each retainer and ignition device assembly, and each initiator assembly shall be radiographically examined for defects as specified in 4.4.3.8.

4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)

4.2 Inspection conditions. Inspections shall be in accordance with MIL-A-2550.

4.3 First article sample.

4.3.1 Initial production sample. At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 21 assembled initiators for ballistic tests, a quantity of ignition devices in accordance with 4.3.2 assembled and crimped into the retainer, 9 sets of unassembled components, one set of unplated components and 50 shear pins from the same lot of shear pins tested by the contractor. The sample shall be subjected to the tests of 4.4.3. Test procedures shall be in accordance with 4.5. The sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.3.1.1 Submission. First article samples and ballistic acceptance test samples of ballistic assemblies shall be in accordance with their applicable specification.

4.3.1.2 Examinations and tests. After examination and provisional acceptance at the source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.3.1.3 Initial production failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3.2 Sample size for sensitivity tests. The number of ignition devices, when assembled and crimped into the retainer, submitted for ignition device sensitivity tests shall be in accordance with table II.

4.4 Inspection provisions.

4.4.1 Lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.4.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with ANSI/ASQ Z1.4.

TABLE II. Sensitivity test sample size.

Lot Size	Ignition Device Sensitivity Sample Size
3 to 15	6
16 to 25	8
26 to 40	10
41 to 65	14
66 to 110	20
111 to 180	30
181 to 300	50
301 to 500	60
501 to 800	70
801 to 1300	80
over 1300	100

4.4.1.2 Size of lot. A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with acceptable explosive and pyrotechnic components.

4.4.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burrs and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gauged to determine applicability. All nonconforming material shall be rejected.

4.4.2.1 Pin, Firing, Drawing 11738410, covering a detail of Drawing 11731492.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	True position of O-ring groove with body	Gage
102	Concentricity of striker with body	Gage
103	True position of shear pin hole with body	Gage
104	Diameter of O-ring groove, min	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

105	Diameter of body, min	Gage
106	Diameter of striker	Gage
107	Length of striker, min	Gage
108	Diameter of shear pin hole, max	Gage
109	Depth of shear pin hole, max	Gage
110	Width of O-ring groove	Gage
111	Location of shear pin hole	Gage
112	Permissible variation of O-ring groove	Gage
113	Spherical radius of tip	Gage
114	Metal defective	Visual
115	Protective finish inadequate or defective	Visual
116	Burr or sharp corner, except where permitted	Visual
117	Foreign matter	Visual
Minor		
201	Diameter of O-ring groove, max	Gage
202	Diameter of body, max	Gage
203	Diameter of shear pin hole, min	Gage
204	Depth of shear pin hole, min	Gage
205	Width of slot	Gage
206	Depth of slot	Gage
207	Total length	Gage
208	Length of body to O-ring groove	Gage
209	Surface finish improper, except as otherwise classified	Visual

4.4.2.2 Housing, Drawing 11731622, covering a detail of Drawing 11731492.

Critical		
1	Surface finish of cavity	Gage
Major		
101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Pitch diameter of internal thread, max	Gage
104	Minor diameter of internal thread, max	Gage
105	True position of pitch diameter of external thread with end flange diameter	Gage
106	Concentricity of pitch diameter of internal thread with large end of center cavity	Gage
107	True position of pitch diameter of internal thread with center cavity	Gage
108	True position of striker hole with center cavity	Gage
109	True position of shear pin hole with center cavity	Gage
110	True position of flats with internal thread	Gage
111	Perpendicularity of striker face with center cavity	Gage
112	Perpendicularity of striker face with internal thread on large end	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

113	Diameter of shear pin hole	Gage
114	Diameter of shear pin hole counterbore	Gage
115	Diameter of striker hole	Gage
116	Diameter of bore, large end	Gage
117	Depth of bore, large end	Gage
118	Diameter of bore, small end	Gage
119	Depth of bore, small end	Gage
120	Depth of shear pin counterbore hole	Gage
121	Length of shear pin hole	Gage
122	Length of internal thread, on large end	Gage
123	Length of internal thread on small end, min	Gage
124	Length of internal thread on small end, max	Gage
125	Width of thread undercut	Gage
126	Location of shear pin hole	Gage
127	Diameter of center cavity	Gage
128	Width of striker plate	Gage
129	Diameter of counterbore, small end	Gage
130	Chamfer at mouth of small end	Gage
131	Thread lead angle	Gage
132	Protective finish inadequate or defective	Visual
133	Burr or sharp corner	Visual
134	Foreign material	Visual
135	Metal defective	Visual/X-ray
Minor		
201	Pitch diameter of external thread, max	Gage
202	Pitch diameter of internal thread, min	Gage
203	Width across flats	Gage
204	Diameter of flange	Gage
205	Width of flange	Gage
206	Depth of counterbore	Gage
207	Location of external thread from face	Gage
208	Location of shoulder from face	Gage
209	Width of shoulder	Gage
210	Diameter of shoulder	Gage
211	Total length	Gage
212	Diameter of large end	Gage
213	Angle at bottom of bore, small end	Gage
214	Surface finish improper, except as otherwise classified	Visual

4.4.2.3 Pin, Shear, Drawing 11731626, covering a detail of Drawing 11731492.

Critical		
1	Metal defective	Visual
Major		
101	Diameter, min	Gage
102	Total length, min	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

103	Surface finish improper	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual

Minor

201	Diameter, max	Gage
202	Total length	Gage

4.4.2.4 Bracket, Mounting, Drawing 11731624, covering a detail of Drawing 11731474.

Critical
None defined

Major

101	Pitch diameter of internal thread, max	Gage
102	Minor diameter of internal thread, max	Gage
103	Diameter of mounting holes	Gage
104	Length of mounting holes centers	Gage
105	Height from base to thread hole center	Gage
106	Length from shoulder to thread hole center	Gage
107	Shoulder radius incorrect	Gage
108	Width of shoulder	Gage
109	Protective finish inadequate or defective	Visual
110	Burr or sharp corner	Visual
111	Foreign matter	Visual

Minor

201	Pitch diameter of internal thread, min	Gage
202	Major diameter of internal thread, min	Gage
203	Total base length	Gage
204	Width of mounting surface (base)	Gage
205	Height from base to top of shoulder	Gage
206	Thickness of base	Gage
207	Length from center line to base end	Gage
208	Mounting hole location	Gage
209	Length of base relief	Gage
210	Depth of base relief	Gage
211	Base relief location incorrect	Gage
212	Angle of base relief incorrect	Gage
213	Radius, angles, or chamfers missing or incorrect	Gage
214	Surface finish improper	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.4.2.5 Ignition Device (Percussion), PVU-1/A, Drawing 851AS110, covering a detail of Drawing 11731585.

Critical		
1	Improper assembly	Visual/X-ray[1/]
2	Any part missing	Visual/X-ray[1/]
3	Pellet broken, cracked or dislodged	Visual
4	Metal defective	Visual
5	Foil discolored	Visual
6	Mixed ignition devices	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		
None defined		

[1/] The ignition devices shall be X-rayed before assembly into the retainer. The X-ray negatives, properly assembly, shall be forwarded to the procuring activity.

4.4.2.6 Ignition Device, Retainer, Drawing 11731620, covering a detail of Drawing 11731585.

Critical		
1	Diameter of ignition device	Gage
2	Depth of ignition device pocket	Gage
3	Metal defective	Visual
Major		
101	Outside diameter	Gage
102	True position of small inside diameter with outside diameter	Gage
103	True position of large inside diameter with outside diameter	Gage
104	Protective finish inadequate or defective	Visual
105	Surface finish of ignition device pocket improper	Visual
106	Burr or sharp corner	Visual
107	Foreign matter	Visual
Minor		
201	Diameter of small inside diameter	Gage
202	Total length	Gage
203	Radius or chamfer missing or incorrect	Visual
204	Surface finish improper, except as otherwise classified	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.4.2.7 Retainer and Ignition Device Assembly, Drawing 11731585, covering a detail of Drawing 11738753.

Critical		
1	Ignition device not properly seated	Visual
Major		
101	Ignition device not properly sealed and crimped	Visual
Minor		
None defined		

4.4.2.8 Delay Element Assembly, 0.30 Second, Delay, Drawing 11738753, covering a detail of Drawing 11731474.

Critical		
1	Improper assembly	Visual/X-ray[1/]
2	Any part missing	Visual/X-ray[1/]
3	Mixed ammunition	Visual
4	Flash hole missing or obstructed	Visual
5	Marking missing, incorrect, or illegible	Visual

[1/] The delay element shall be X-rayed as part of the final assembly X-ray.

Major		
101	Output charge not visible through flash hole	Visual
102	Ignition device retainer not flush or below surface "S"	Visual
103	Junction by ignition device retainer and delay body not properly sealed or crimped	Visual
104	Protective finish inadequate or defective	Visual
105	Metal defective	Visual/X-ray
Minor		
201	Varnish missing	Visual

4.4.2.9 Initiator M104 Assembly, Drawing 10551446, covering a detail of Drawing 11731474.

Critical		
1	Mixed ammunition	Visual
2	Metal defective	Visual
3	Vent hole obstructed	Visual
4	Marking missing, incorrect, or illegible	Visual
Major		
101	Thread damaged	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

102	Burr	Visual
103	Foreign matter	Visual

Minor
None defined

4.4.2.10 Firing Mechanism Assembly, Drawing 117311492, covering a detail of Drawing 11731474.

Critical		
1	Firing pin protrusion	Gage
2	Shear pin not engaged in firing pin	Gage
3	Striker hole in initiator housing obstructed	Visual
Major		
101	O-ring not lubricated	Visual
102	Burr	Visual
103	Foreign matter	Visual

Minor
None defined

4.4.2.11 Assembly, Drawing 11731474.

Critical		
1	Any part missing	Visual and X-ray
2	Improper assembly	Visual and X-ray
3	Total length, max	Gage
4	Marking missing, incorrect, or illegible	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		
201	Thread sealing compound missing	Visual
202	Protective cap or plug cap missing	Visual
203	Protective finish inadequate or defective	Visual

4.4.2.12 Box, Fiberboard (Inner) Packing, Ammunition for Initiator, Delay, Drawing 9280891.

Critical
None defined

Major
None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Sealing strip missing or improperly applied	Visual
205	Cut, damaged, or improperly sealed bag	Visual

4.4.2.13 Box, Packaging, Ammunition for Initiator, Delay, Drawing 9280892.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.4.2.14 Box, Fiberboard (Outer) Alternative, Ammunition Packing, Drawing 9280893.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual
202	Carton broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.4.3 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.3.1 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with the requirements of 3.11 shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.4.3.2 Protective finish. Unless otherwise specified, sampling plans and procedures in determination of acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in ANSI/ASQ Z1.4 and MIL-A-8625.

4.4.3.3 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.6 shall be cause for rejection of the assembly.

4.4.3.4 Breakaway torque. Each initiator shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirements of 3.5 shall be cause for rejection of the assembly.

4.4.3.5 Leak rate. Each initiator, less the firing mechanism and bracket, shall be subjected to a leak rate test. Failure of any initiator to comply with the requirements of 3.8 shall be cause for rejection of the initiator. This test shall be done by the contractor before final assembly.

4.4.3.6 Ignition device sensitivity. A sample of ignition devices in accordance with table II, when assembled and crimped into the retainer, form the initial production sample. All lots produced after this initial production sample shall be tested for sensitivity.

4.4.3.6.1 All fire. One half the ignition device sample size, crimped into the retainer, shall be tested at the 26.0 inch ounce energy level in accordance with 3.4.1. Any ignition device failing to comply with 3.4.1 shall be cause for rejection.

4.4.3.6.2 No-fire. One half the ignition device sample size, crimped into the retainer, shall be tested at the 4.0 inch ounce energy level in accordance with 3.4.1. Any ignition device failing to comply with 3.4.1 shall be cause for rejection.

4.4.3.7 Functional. The following tests shall be performed by the Government.

4.4.3.7.1 Ballistics. Twenty-one units shall be tested to determine compliance with the requirements of 3.7.1, 3.7.2, and 3.7.3. The units shall be temperature conditioned as provided in table III and 4.5.7.1.1. Failure of any unit to comply with the requirements of 3.7.1, 3.7.2, and 3.7.3 shall be cause for rejection of the lot.

TABLE III. Sample size for ballistic temperature conditioning.

Temperature	Sample Size
-65 to -70°F	7
65 to 75°F	7
200 to 205°F	7

4.4.3.8 Radiographic examination. All ignition device assemblies, retainer and ignition device assemblies, and initiator assemblies shall be radiographically examined for defects.

4.4.3.8.1 Ignition device, PVU-1/A, Drawing 851AS110. Prior to assembly into the cartridge, the ignition device shall be radiographically examined in accordance with ASTM E 1742. Each ignition device shall be X-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

rayed on boards or trays with the cup side perpendicular to the rays. The X-ray negatives shall be identified by item nomenclature and lot number. Ignition devices are to be left mounted to the boards or trays until after the X-rays have been reviewed and defects identified to facilitate removal of defectives. After screening, the X-rays with defectives identified shall be forwarded to the engineering agency designated by the procuring activity. Film interpretation of the radiographs shall identify any of the following: missing or abnormally thin charge; charge broken, cracked, or porous; cracked or off-center anvil or other misplaced or deformed inert parts; anvil abnormally close to the ignition device cup dome; and all other observable defects in assembly.

4.4.3.8.2 Retainer and ignition device assembly, Drawing 11731585. All retainer and ignition device assemblies shall be radiographically examined in accordance with ASTM E 1742, with the exception that the retainer and ignition device assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The retainer and ignition device assemblies shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All retainer and ignition device assemblies shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing serial numbers are to be identified on the X-rays. The retainer and ignition device assemblies shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic identification shall include item nomenclature, the complete lot number assigned to that lot, and the span of serial numbers displayed. Radiographs of the entire production lot shall be submitted to the activity conducting the production lot acceptance tests. Defective units found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Any assembling of the retainer and ignition device assembly into initiators prior to notification by the contracting agency of X-ray acceptance for the retainer and ignition device assembly shall be at the contractor's risk. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

1. Ignition charge present and not cracked
2. Anvil present, seated, and not cocked
3. No material defects

4.4.3.8.3 Initiator assembly, Drawing 11731474. All assembled initiators shall be radiographically examined in accordance with ASTM E 1742, with the exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiator shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initiators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing serial numbers are to be identified on the X-rays. The initiators shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic identification shall include item nomenclature, the complete lot number assigned to that lot, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance tests. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct position of the following parts:

a. Ignition Device Retainer	P/N 11731620
b. Ignition Device	P/N 851AS110
c. M104 Assembly	P/N 10551446

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

d. Propellant	MIL-P-46994
e. Housing	P/N 11731622
f. Shear Pin	P/N 11731626
g. Firing Pin	P/N 11738410
h. O-ring	MS28775-009
i. Mounting Bracket	P/N 11731624
j. Delay Element Body	P/N 11731623

4.4.4 Inspection equipment. The examination and tests shall be made using equipment approved by the government.

4.5 Test method and procedures.

4.5.1 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear strength test to determine compliance with 3.11. The test shall be performed in the fixture in conjunction with a tensile testing machine.

4.5.2 Protective finish. Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.5.3 Breakaway torque. A torque wrench set at the torque requirement specified in 3.5 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torque shall be applied (1) between the female body and male body, (2) between the male or female body and the delay element body, (3) between the delay element body and the firing mechanism assembly, and (4) between the mounting bracket and the firing mechanism.

4.5.4 Residual magnetism. The initiator shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects, by placing the initiator 5 inches from and at the same height as the compass in a north-south horizontal position (firing mechanism end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degrees of rotation of the initiator.

4.5.5 Leak rate. The method and procedure for performing the leak rate test shall be as approved by the Government.

4.5.6 Ignition device sensitivity. Drop test the sample of ignition devices in accordance with table II of 4.3.2 assembled and crimped into the retainer from the initial production sample and a sample of primers in accordance with table II of 4.3.2 assembled and crimped into the retainer from each production lot. One-half of the sample at 13 inches and the other half at 2 inches using a steel ball 15/16 inches in diameter, and weight 1.94 ± 0.02 ounces in a primer sensitivity machine with a holder approved by the procuring activity. The heights of 13 and 2 inches shall be adjusted taking into consideration the actual weight of the ball to meet the energy levels specified in 3.4.1.

4.5.7 Functional.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.5.7.1 Ballistics. The test sample shall be rigidly supported in the fixture and shall be tested within 5 minutes after its removal from the conditioning chamber. When this time limit is exceeded the sample shall be reconditioned in accordance with 4.5.7.1.1.

4.5.7.1.1 Temperature conditioning. A standard M113 Initiator loaded with dummy IIA pellets and a dummy PVU-1/A Ignition Device and delay element containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems. The initiator was designed for use in military aircraft, and has no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced .

c. Production test activity and lot size (see 4.1 and 4.3.1).

d. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.4.3.8.3).

e. Level of packing and marking (see 5.1, 6.5).

f. That the safety precaution requirements of the ``Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

activity, the safety precaution requirements of ``Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Hazard notice. The initiator described herein and certain of its components are flammable, toxic, or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or sample selected from the first production items , and the number of items to tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.5.1 Packing and marking. Packing and Marking should be Level A or B, as specified in the contract.

6.5.1.1 Level A. Level A packing and marking should be in accordance with Drawings 9280891 and 9280892.

6.5.1.2 Level B. Level B packing and marking should be in accordance with Drawings 9280891 and 9280893.

6.5.1.3 Air Force. Air Force deliveries should be in accordance with TPO F00-138-3770.

6.5.1.4 Lot numbers. Lot numbering should be in accordance with MIL-STD-1168.

6.6 Definitions.

6.6.1 Ignition delay. Ignition delay is defined as the time that elapses between the firing of the delay element and the beginning of the continuous pressure rise on the pressure-time record.

6.6.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

6.7 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.8 Single contractor. If a single contractor manufactures the total item, the classification of defects does not apply. The appropriate specifications will apply.

6.9 Initiator component. When the M104 initiator is being manufactured as a component of the M113 initiator those tests such as residual magnetism and radiographic inspection specified for the M104 in MIL-DTL-50374 and MIL-DTL-50375 which are also required by this specification are waived for the component items but shall be done on the assembled M113 initiator as specified herein.

6.10 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. This revision was authorized by ECP number 045220I016.

Preparing activity
Navy - OS
(Project 1377-0221)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.